



KIRLOSKAR CHILLERS PRIVATE LIMITED

A Kirloskar Group Company

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Document Name : Sound Pressure Levels of all Screw Chiller Series				
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Engineering Specification

This specification gives information of Screw Chillers sound pressure levels as per ANSI/AHRI Standard 575 (Method of Measuring Machinery Sound within an Equipment Space).

Table A- Sound Pressure Level (dB) at 100% Load at various distances in meter from end of the unit:

Chiller Series	Sound Pressure Level (dB) at 100% Load and at various distances			
	1 m	3 m	5 m	10 m
KWE Single Circuit	81.0	77.0	76.0	75.0
KWE Dual Circuit	82.0	78.0	77.0	76.0
KWK Single Circuit	81.0	77.0	76.0	75.0
KWK Dual Circuit	82.0	78.0	77.0	76.0
KWI Single Circuit	80.0	76.0	75.0	74.0
KWI Dual Circuit	81.0	77.0	76.0	75.0
KWS Single Circuit	82.0	78.0	77.0	76.0
KWS Dual Circuit	83.0	79.0	78.0	77.0
KAA/KAS Single Circuit	82.0	78.0	77.0	76.0
KAA/KAS Dual Circuit	83.0	79.0	78.0	77.0
KAA/KAS Triple Circuit	85.0	81.0	80.0	79.0

Table B- Sound Pressure Level (dB) at 1 meter distance from the end of unit at various part load conditions:

Chiller Series	Sound Pressure Level (dB) at various Part Loads at 1m distance			
	100%	75%	50%	25%
KWK Single Circuit	81.0	78.0	79.0	80.0
KWK Dual Circuit	82.0	79.0	80.0	81.0
KWI Single Circuit	80.0	77.0	78.0	79.0
KWI Dual Circuit	81.0	78.0	79.0	80.0
KWS Single Circuit	82.0	79.0	80.0	81.0
KWS Dual Circuit	83.0	80.0	81.0	82.0



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KAA/KAS Single Circuit	82.0	79.0	80.0	81.0
KAA/KAS Dual Circuit	83.0	80.0	81.0	82.0
KAA/KAS Triple Circuit	85.0	82.0	83.0	84.0

Table C- Sound Pressure Level (dB) at 1 m distance from the end of unit at 100% Load Condition and at various octave bands:

Chiller Model	Sound Pressure Levels at Various Octave Bands at 100% Load and at 1m distance from Chiller Unit									A
	32	63	125	250	500	1000	2000	4000	8000	
KWE Single Circuit	23.1	35.7	47.8	58.9	75.3	75.1	70.9	55.9	48.2	80.9
KWE Dual Circuit	20.3	33.6	49.5	52.5	79.8	77.3	71.6	56.4	42.2	82.2
KWK Single Circuit	28.1	42.3	56.6	67.1	68.9	71.2	76.4	77.2	63.7	81.1
KWK Dual Circuit	24.2	43.3	54.4	66.7	69.2	72.2	77.1	79.1	64.1	82.1
KWI Single Circuit	19.8	34.2	50.9	58.9	73.3	75.4	71.9	56.3	46.6	80.2
KWI Dual Circuit	29.1	48.5	57.3	66.5	70.4	73.5	75.9	76.8	62.4	81.1
KWS Single Circuit	21.3	36.2	47.7	62.5	70.7	81.6	69.4	56.3	44.3	82.2
KWS Dual Circuit	26.6	50.1	57.2	66.1	68.7	71.6	77.5	80.5	66.4	83.1
KAA/KAS Single Circuit	28.3	45.1	56.5	65.7	68.3	73.7	78.8	77.8	63.2	82.4
KAA/KAS Dual Circuit	26.6	50.1	57.2	66.1	68.7	71.6	77.5	80.5	66.4	83.1
KAA/KAS Triple Circuit	28.5	42.4	58.1	69.8	69.7	74.4	79.4	82.5	67.6	85.1

Notes:

1. Sound measurements are carried out as per ANSI/AHRI Standard 575 – 2008.
2. Above data is for reference purpose to give idea about the sound level in Screw compressor units.
3. Above values are measured without sound attenuation to discharge line and condenser.
4. Noise levels mentioned here are considering free field atmosphere.
5. We can reduce KAS/KAA chiller noise by 10 dB by providing Acoustic enclosure to compressor provided atmospheric noise should not be more than 55 dB.